

Python Basics

Python is a high-level, general-purpose programming language created by Guido van Rossum and first released in 1991. It emphasises code readability and uses indentation to define code blocks rather than curly braces.

Python supports multiple programming paradigms including procedural, object-oriented, and functional programming. Its large standard library and active ecosystem of third-party packages make it popular for web development, data science, and automation.

Python Data Structures

Python has four built-in collection types: lists, tuples, sets, and dictionaries.

A list is an ordered, mutable sequence: [1, 2, 3]. A tuple is an ordered, immutable sequence: (1, 2, 3). A set is an unordered collection of unique elements: {1, 2, 3}. A dictionary maps keys to values: {'name': 'Alice', 'age': 30}.

Lists are the most flexible; tuples are faster and used for fixed data; dictionaries are the go-to for fast key-based lookup.

Python Functions and Scope

Functions in Python are defined with the `def` keyword. They support default arguments, keyword arguments, and variable-length argument lists via `*args` and `**kwargs`.

Python uses LEGB scope resolution: Local, Enclosing, Global, Built-in. A variable is looked up in the local scope first, then in any enclosing function scopes, then in the module global scope, and finally in the built-in namespace.